decreases.

21 From $R = \dfrac{\rho L}{A}$, Resistance per unit length of each wire,

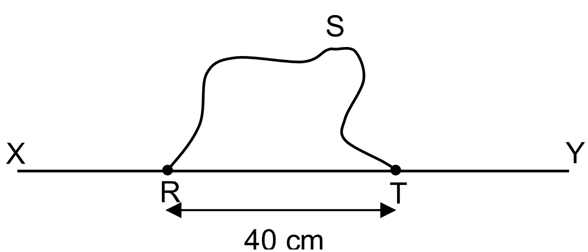
$$\frac{R}{L} = \frac{\rho}{A} = \frac{1.6 \times 10^{-6}}{\pi\left(\dfrac{0.0016}{2}\right)^2} = 0.80\ \Omega\ \text{m}^{-1}$$

Resistance between XR, RT, and TY (each 4.0 m)

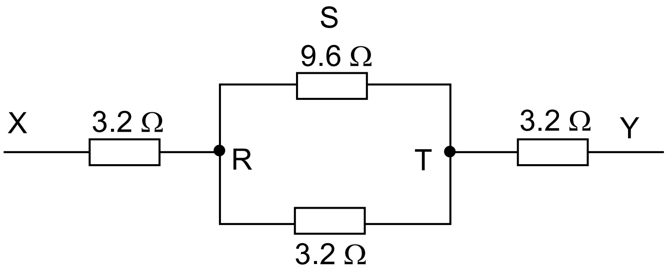
$= \dfrac{R}{L} \times L = 0.80 \times 4.0 = 3.2\ \Omega$

Resistance between RST (12.0 m)

$= \dfrac{R}{L} \times L = 0.80 \times 12.0 = 9.6\ \Omega$



Hence equivalent circuit looks like this:



$$R_{effective} = 3.2 + \left(\frac{1}{9.6} + \frac{1}{3.2} \right)^{-1} + 3.2 = 8.8\ \Omega$$

22 Resistance of thermistor increases when temperature of thermistor is reduced